

RESEARCH PROTOCOL

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1. Title

Intelligent Tutor Based on Multi-Agent Architecture and Language Models for Adaptive Learning of Ancash Quechua

2. Summary

This project proposes the design and implementation of an intelligent tutor based on a multi-agent architecture and language models with retrieval augmented by retrieval (RAG) for the adaptive learning of Ancash Quechua, a language considered to have limited digital resources. The research adopts the Design Science Research paradigm, developing a technological artifact that integrates specialized agents, semantic retrieval, and natural language processing to provide personalized feedback. Subsequently, the system will be evaluated through a pilot study using metrics of linguistic accuracy, usability (SUS), and learning gain (Vaswani et al., 2017; Zheng et al., 2023).

Keywords: Ancashino Quechua, Intelligent Tutor, Multiagent, RAG, LLM.

3. Introduction and problem statement

Ancash Quechua is one of the most important linguistic varieties in Peru; however, it faces a progressive process of linguistic displacement due to the predominance of Spanish in educational and technological contexts (Ccaccachahua Huamaní, 2022).

Recent advances in Intelligent Tutoring Systems (ITS) and Large Scale Language Models (LLMs) have shown significant improvements in language teaching with abundant digital resources, although their performance decreases considerably when processing morphologically complex languages with few available corpora (Zheng et al., 2023).

Furthermore, research on automatic processing of Quechua shows that the main limitation continues to be the scarcity of annotated linguistic data, a situation that can be mitigated through augmented retrieval architectures (ARA) and specialized models (Ríos & Castro, 2024; Gonzales & Castro-Mamani, 2014).

In this context, the following research question arises:

How does the design and implementation of an Intelligent Tutor based on a multi-agent architecture allow for the optimization of adaptive learning of Ancash Quechua in university and continuing education students?

Justification

Theoretical justification

The study contributes to the development of Natural Language Processing for low-resource languages, integrating generative models with semantic retrieval mechanisms (Ríos & Castro, 2024).

Practical justification

It will allow for an accessible conversational environment for students, teachers and researchers interested in learning Ancashino Quechua (Ccaccachahua Huamaní, 2022).

Technological justification

The proposal demonstrates the applicability of modern intelligent agent architectures using tools such as LangChain and LangGraph to build specialized conversational systems (LangChain, n.d.).

4. Literature review

Recent literature shows an evolution from traditional chatbots to multi-agent architectures where different specialized agents collaborate during the teaching process (Zheng et al., 2023).

In the field of Quechua, several works highlight the need to combine rule-based linguistic approaches with probabilistic models due to the complexity of its agglutinative morphology (Gonzales & Castro-Mamani, 2014).

Furthermore, learning transfer and enhanced retrieval strategies have been shown to improve the accuracy of models applied to Quechua Ancashino (Ríos & Castro, 2024).

Theoretical foundations

Intelligent Tutoring Systems (ITS)

ITS structures learning through student, domain, pedagogical, and interface models (Zheng et al., 2023).

Multi-agent Systems

Multi-agent systems allow the distribution of functions among specialized agents coordinated by a supervisory agent, reducing the complexity of conversational reasoning (LangChain, n.d.).

Natural Language Processing for agglutinative languages

The suffixal structure of Ancash Quechua requires specific tokenization techniques and morphological validation (Gonzales & Castro-Mamani, 2014).

RAG Architecture

Retrieval Augmented Generation improves the accuracy of models by querying a knowledge base before generating answers (LangChain, n.d.).

Transformers

Transformer architecture forms the basis of modern language models through auto-regressive attention mechanisms (Vaswani et al., 2017).

5. Questions, hypotheses and objectives

Main question

How does a multi-agent architecture contribute to the adaptive learning of Ancash Quechua?

Hypothesis

Implementing the intelligent tutor based on multi-agent architecture will significantly improve linguistic accuracy, system usability, and learning gain compared to a conventional environment (Zheng et al., 2023).

General objective

Design and implement an intelligent tutor based on multi-agent architecture and adaptive language models.

Specific objectives

- Design the multi-agent architecture.
- Building a linguistic corpus using RAG.
- Develop the web prototype.
- Evaluate linguistic accuracy, usability (SUS), and learnability.

6. Methodology

Paradigm

The research is developed under the Design Science Research paradigm, oriented towards the design, construction and evaluation of a technological artifact.

Approach

Mixed with quantitative predominance.

Method

Development and evaluation of an intelligent system.

Phases

1. Collection and validation of the linguistic corpus.
2. Multi-agent architecture design.
3. Implementation using FastAPI, LangChain and LangGraph (LangChain, sf).
4. RAG system integration.
5. Pilot evaluation.

Tools

- Pretest-posttest test.
- SUS scale.
- Record of interactions.
- Linguistic accuracy metrics.

7. Ethical considerations

Informed consent, anonymization of personal data, and responsible use of artificial intelligence will be applied. Furthermore, the project will respect the linguistic and cultural heritage of Ancash Quechua, following documented principles of cultural preservation for indigenous languages (Ccaccachahua Huamani, 2022).

8. Expected Results

Expected results:

- Functional multi-agent architecture.
- Validated linguistic corpus.
- Operational web prototype.
- Increase in linguistic precision.
- High system usability (SUS).
- Evidence of improvement in learning (Zheng et al., 2023).

9. Schedule and budget

Activity	M1	M2	M3	M4
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Literature review	X			
Architectural design	X	X		
Prototype development		X	X	
RAG Integration			X	
Pilot evaluation			X	X
Final draft				X

Reference budget: software development, cloud infrastructure, vector storage, user validation, and scientific dissemination.

10. Bibliography

- Ccaccachahua Huamaní, B. (2022). Préstamos léxicos en el campo de las tecnologías modernas del castellano en el quechua chanca del anexo de Puncuhuacca. *Lengua y Sociedad*, 21(2), 449–474. <https://doi.org/10.15381/lengsoc.v21i2.22799>
- Gonzales, A. R., & Castro-Mamani, R. A. (2014). Morphological disambiguation and text normalization for southern Quechua varieties. In *Proceedings of the First Workshop on Applying NLP Tools to Similar Languages, Varieties and Dialects* (pp. 39–47).
- LangChain. (s.f.). LangChain overview. <https://langchain-5e9cc07a.mintlify.app/oss/python/langchain/overview>
- Ríos, G., & Castro, M. (2024). Estrategias de Transferencia de Aprendizaje para el Procesamiento del Lenguaje Natural en Quechua Ancashino. *Anales Científicos UNMSM*, 28(1), 112–126.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems*, 30.
- Zheng, Y., Zhang, L., & Wang, X. (2023). Multi-agent conversational tutors for low-resource language acquisition: Architecture and pedagogical evaluation. *Computers & Education: Artificial Intelligence*, 4, 100120.